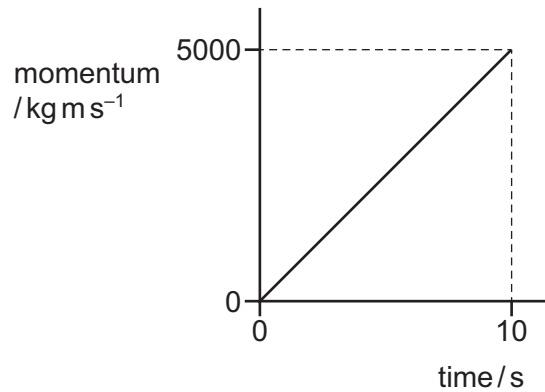


- 11 The graph shows how the momentum of a motorcycle changes with time.



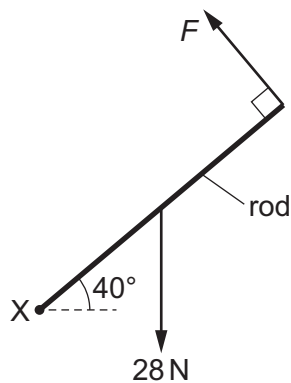
What is the resultant force on the motorcycle?

- A** 500 N **B** 5000 N **C** 25 000 N **D** 50 000 N
- 12 A rocket has a weight of W and an initial acceleration of a as the rocket leaves the ground vertically. The acceleration of free fall is g .

Which expression gives the initial upward force exerted on the rocket due to the engine?

- A** $\frac{Wa}{g} + W$ **B** $\frac{Wa}{g}$ **C** $\frac{Wa}{g} - W$ **D** $Wa + W$
- 13 A uniform rod of weight 28 N is supported at one end by force F .

The rod is in equilibrium and attached to a frictionless hinge at end X.



What is the magnitude of F ?

- A** 9.0 N **B** 11 N **C** 18 N **D** 21 N